Cooling Equipment Sensor Fault Detection

A supervised LSTM benchmark on author-labelled air-handling-unit simulation data

Deep Learning Project - Section B
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14 October 2026

Abstract

This project studies outdoor-air temperature sensor-bias detection in an operating simulated air-handling unit (AHU). It compares a supervised LSTM, repeated with three training seeds, against a gradient-boosted classifier using identical sensor histories. Entire seasons separate training, validation and final evaluation. The source authors define the fault labels; the project does not create them from temperature thresholds.

The predeclared primary LSTM achieves 92.82% accuracy, 99.74% precision, 91.86% recall and F1 95.64% on 33,432 autumn readings. Its 1.44% false-positive rate exceeds the original 1% goal. The 69 false-positive readings form one alarm burst. These results concern one simulated unit and one fault family, not datacenter deployment.

Primary independent-test measure	Value
Accuracy	92.82%
Precision / recall	99.74% / 91.86%
False-positive rate / false bursts	1.44% / 1
Observable fault days detected	35 / 36

The earlier M100 experiments remain supporting research. Their outcome concerns scheduler state and their tested models did not meet the false-alarm targets. The user-approved dataset change provides equipment-fault labels and a more precisely defined task.

1. Problem scope and source data

An AHU conditions and supplies air using fans, dampers, and heating/cooling coils. A biased outdoor-air temperature sensor can affect economizer and coil-control decisions. The binary task is to identify the source-defined sensor-bias condition from current and preceding measurements while the equipment is occupied and running. The study does not forecast future faults or diagnose every HVAC failure.

The public AFDD evaluation collection is hosted by the Open Energy Data Initiative and attributed to Lawrence Berkeley National Laboratory and contributing institutions [1]. The selected MZVAV-1.csv contains 272,160 one-minute records across 189 days. It was generated through EnergyPlus-Dymola co-simulation of a Chicago office-building AHU [2, pp. 3-7].

Item	Reference definition
Outcome	Original Fault Detection Ground Truth: 0 unfaulted; 1 faulted
Fault family	Outdoor-air temperature sensor bias
Severity	+/-1, +/-2 and +/-4 degrees Celsius
Input temperatures	Degrees Fahrenheit; no conversion to Celsius is implied
Equipment scope	One modeled AHU and one simulated building
Sampling	One minute; source local time, timezone unspecified

Every source day label matches the normal/fault schedule in inventory Table 1-2. The target is not reconstructed from the measurements. The rooftop-unit file was unsuitable for this binary study because its native label column contains no normal examples; other collections have fewer independent source days. Dataset selection occurred before model fitting.

Scenario prevalence is experimental. Most provided days contain imposed faults; this is not a measurement of how often equipment fails in service. On the heldout cohort, an all-fault classifier would have 85.71% accuracy but 100% false-positive rate. Accuracy alone is therefore inadequate.

2. Chronological separation and coverage

Partition	Seasons	Raw days	Scored days	Endpoints	Fault endpoints
train	Winter + spring	91	77	61,187	52,446
validation	Summer	49	42	33,432	28,656
test	Autumn	49	42	33,432	28,656

Training uses winter and spring, validation uses summer, and the final test uses autumn. Documented scenario blocks and whole days remain within one partition. The design never randomly splits overlapping windows. No window crosses a day, equipment boundary or timestamp gap.

Eligibility requires all 30 minutes of a history to have occupied mode and the supply fan on. This operational definition was frozen before fitting. Nights, Sundays with no occupied operation, and initial warmup are unscored. The reported point metrics apply only to eligible endpoints; the source labels remain unchanged for all raw days.

Autumn contains 42 raw fault days, of which 36 contain eligible windows. The primary model detects 35 of those 36 observable days, or 35 of all 42 fault days. Both denominators are reported to avoid hiding unobservable conditions. Healthy validation and test each consist of one documented scenario week, which limits evidence about normal-operation variability.

The protocol, source checksums, exact cohorts, dates and scenario membership are saved before training. Date, season, day ID, scenario identity, imposed bias and reference labels are metadata only; they never enter the model feature vector.

Whole-season separation tests a seasonal change for the same simulated unit. It does not establish generalization to a new building or physical equipment. Adjacent minutes and days share conditions, so they are not independent statistical samples.

3. Inputs and causal preprocessing

Input family	Measured information
Temperatures	Supply, supply setpoint, outdoor, mixed and return air
Commands	Supply/return fan speed, outdoor/return dampers, cooling/heating valves
Pressure	Supply duct pressure and pressure setpoint
Derived consistency	Supply error, three air-temperature differences, pressure error, command-weighted mixing residual
Quality information	Thirteen indicators retain original missing measurements

Thirteen raw inputs yield six derived consistency features and thirteen original-missing indicators: 32 channels. The command-weighted residual compares mixed-air temperature with a command-weighted outdoor/return mixture. Damper command is a proxy rather than measured flow fraction; the feature is a heuristic, not a validated thermodynamic calculation.

Only nonfinite values become missing. Negative outdoor Fahrenheit temperatures remain valid. Forward filling uses at most five preceding minutes, resetting at every source day or gap. Remaining missing values use medians fitted on normal active training observations. A missing indicator remains set even when a previous reading or median provides an input value.

StandardScaler fits only normal active winter/spring rows, and values are clipped at ten training standard deviations. Validation and test provide no means, medians or scaling parameters. The original measurements are preserved for replay. Neither unknown labels nor source metadata become numerical feature inputs.

Inference requires 30 consecutive one-minute readings from one occupied, running unit, using the saved feature definitions. The pipeline verifies cadence and day boundaries and applies the saved preprocessing. A CSV can include reference labels, but the scorer ignores them.

4. Deep learning model and baseline

Component	Fixed configuration
History	30 minutes x 32 channels
Encoder	Single LSTM layer, hidden size 32
Classification head	16 units, ReLU, dropout 0.2, one binary logit
Loss	BCEWithLogitsLoss; positive weight = training normal/fault ratio (about 1/6)
Optimization	Adam 0.001; batch 512; gradient norm clip 1
Stopping	Maximum 30 epochs; patience 6 on labelled validation average precision
Repeats	Seeds 7, 42 and 123; seed 42 declared primary before testing

Fault examples outnumber healthy examples. The training positive weight is below one because it balances the loss toward the less frequent healthy class; the experiment does not oversample or rebalance validation/test. Average precision selects a checkpoint, while the alarm policy separately controls the precision/recall/false-alarm tradeoff.

The gradient-boosted classifier receives latest value, trailing mean, standard deviation and slope from the same 30-minute input history. It uses learning rate 0.05, 15 leaves, minimum leaf size 30 and L2 regularization 1. Explicit chronological validation chooses among 50, 100 and 200 iterations. Random internal validation is disabled.

The comparison uses identical endpoints, sensor information and evaluation rules. It distinguishes the value of sequence learning from a strong non-DL baseline. The three seed runs characterize training variation; the highest test-scoring seed is not substituted for the predeclared primary.

5. Validation and frozen alarm policy

Model	AP	Precision	Recall	FPR	Observed fault days
lstm_seed_7	0.9836	100.00%	64.42%	0.00%	36/36
lstm_seed_42	0.9834	100.00%	48.66%	0.00%	36/36
lstm_seed_123	0.9789	100.00%	42.10%	0.00%	36/36
boosting	0.9613	100.00%	57.78%	0.00%	32/36

Validation compares global normal-reading score percentiles (95, 99, 99.5, 99.9, 99.95), PR-derived thresholds and default scores. Immediate, two consecutive highs and two-of-three highs compete. All models use the same 33,432 summer endpoints. The policy target gate requires precision >=30%, recall >=40%, FPR <=1%, false bursts <=0.1 per scored device-day, and observable fault-day recall >=60%.

Among feasible policies, selection maximizes precision, then F0.5, then minimizes false-burst rate, then maximizes recall. All three LSTM seeds meet this validation gate. The fixed primary uses logit threshold 6.967112 with two high scores in three consecutive scored minutes; state resets at day boundaries and gaps.

The seed-42 validation policy has 100% empirical precision and 48.66% reading recall. This operating point favors avoiding false alarms. It does not imply every fault reading is detected or that the score is a calibrated fault probability. A single uploaded window supplies one endpoint and cannot establish persistence.

Model, preprocessing and policy hashes were frozen before autumn evaluation. The evaluation records all three neural seeds and the tree. It returns the saved final results on rerun; no threshold, feature, seed or architecture is adjusted after seeing the test season.

6. Independent autumn results

Measure	Primary seed-42 result
Accuracy / F1	92.82% / 95.64%
Precision / recall	99.74% / 91.86%
AP / ROC-AUC	0.9962 / 0.9852
FPR / false discovery ratio	1.44% / 0.26%
False readings / bursts	69 readings / 1 burst
False bursts per scored device-day	0.0431
Fault days	35/36 observable; 35/42 total

Author label / prediction	Healthy	Fault
Healthy	4,707	69
Fault	2,332	26,324

False-positive rate is FP/(FP+TN); false discovery ratio is FP/(TP+FP). The denominators differ. The 1.44% FPR exceeds the 1% goal, so the project does not claim that every engineering target was met. One false alarm burst meets the false-burst budget. All figures retain the frozen validation policy.

The LSTM detects 91.86% of fault readings. The tree detects 72.05% with zero test false positives. The LSTM therefore increases sensitivity with a small false-alarm cost. This evidence concerns a simulated, fault-heavy experiment; comparisons with M100 rare scheduler-state labels are not direct performance improvements on the same population.

7. Seed variation and fault severity

Model	Accuracy	Precision	Recall	FPR
lstm_seed_7	96.66%	99.73%	96.36%	1.55%
lstm_seed_42	92.82%	99.74%	91.86%	1.44%
lstm_seed_123	85.55%	99.71%	83.39%	1.47%
boosting	76.04%	100.00%	72.05%	0.00%

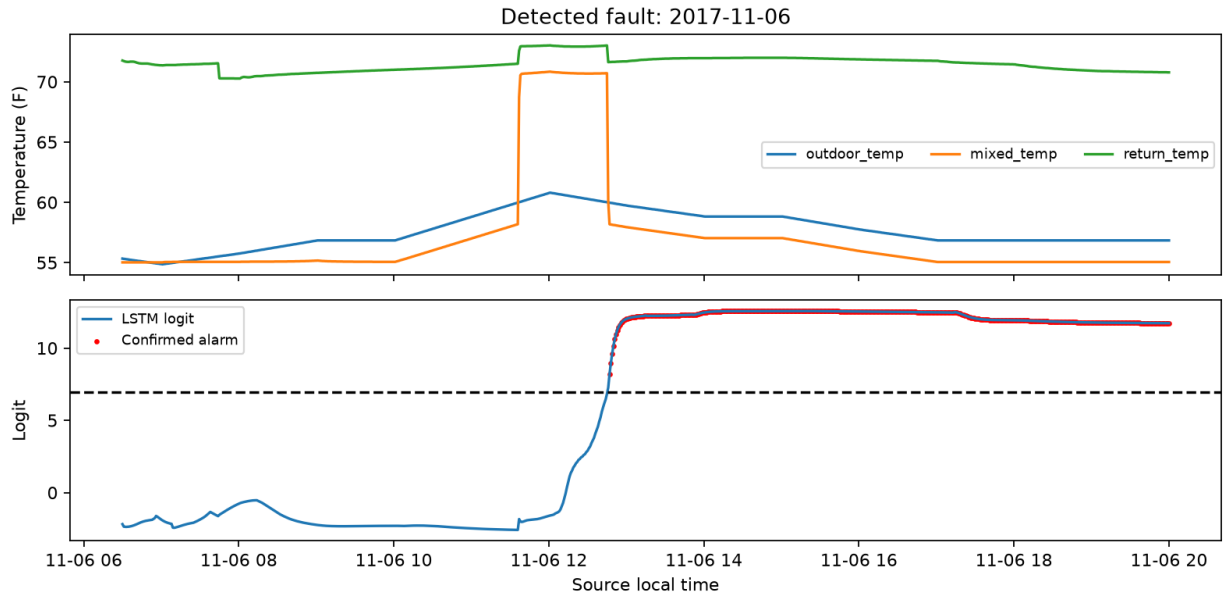
Three-seed test accuracy mean is 91.68% with sample standard deviation 5.64%. This describes variation across three training seeds, not a confidence interval for future equipment. Seed 7 scores higher on test, but seed 42 remains the primary by the predeclared rule.

Bias (C)	Scored days	Reading recall	Detected days
-4	6	99.75%	6
-2	6	96.11%	6
-1	6	78.50%	5
+0	6	Healthy	0
+1	6	87.44%	6
+2	6	90.75%	6
+4	6	98.64%	6

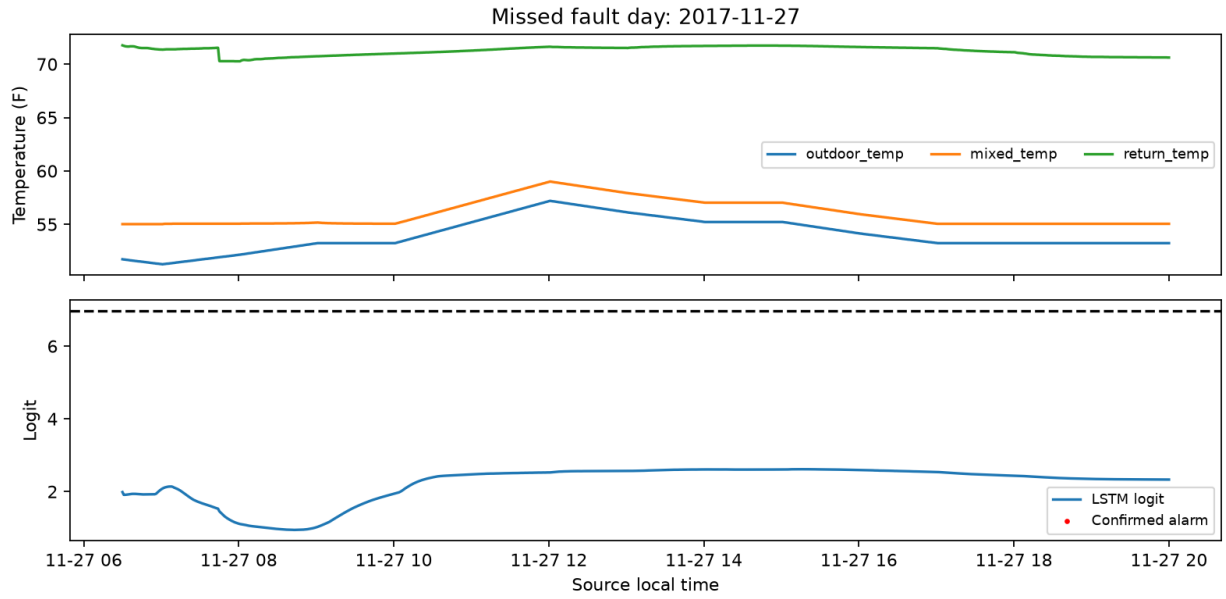
The -1 degree Celsius cases have the lowest reading recall (78.50%) and contain the one observable missed fault day. Larger biases are easier for the frozen model in this test, but the experiment does not establish causal attribution to an individual feature or universal severity monotonicity.

Median delay from source midnight fault imposition is 392 minutes. That includes overnight nonoperation and the 30-minute history requirement. Median delay after the first eligible score is 2 minutes. Reporting only the latter would hide coverage and warmup; neither timing measure establishes early warning.

8. Detection and missed-condition examples

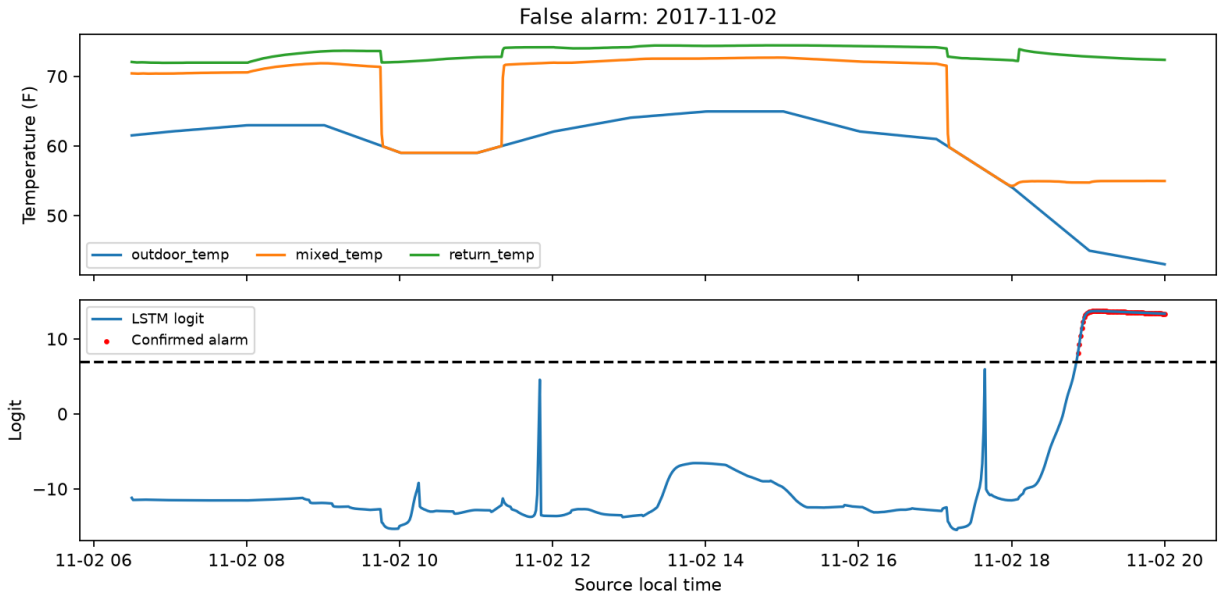


Detected example: 2017-11-06, imposed bias +1 degrees Celsius. The raw temperatures and frozen logit trace show when confirmed alarms occur.



Missed observable day: 2017-11-27, bias -1 degrees Celsius. This is a retained failure case, not excluded from evaluation.

9. False alarm and study limits



False-alarm example: 2017-11-02, an author-labelled healthy condition. The 69 false readings form one burst. The threshold is not raised after seeing this case.

Only one simulated unit, one fault family and specified operating mode are tested. The study does not validate a deployed datacenter monitor, predict future failures or demonstrate general equipment-fault diagnosis. Experimental prevalence is not operational prevalence.

Normal validation/test each cover one scenario week, and adjacent minutes/days are correlated. A binomial interval over thousands of overlapping readings would exaggerate independence. Seed variation and per-day counts are reported, but no precise field-reliability confidence interval is claimed.

Future work should use separate physical units and more fault families, reserve new evaluation periods, and assess calibration, startup coverage and real operational class prevalence. Further development must not tune against the already-inspected autumn holdout.

10. Reproducibility, contribution and references

The GitHub project includes trusted trained artifacts, the source archive/inventory, dashboard replay, CSV inference examples, notebook, model card and reproducible scripts. Verification reproduces all validation/test score arrays, 120 validation policies, source-label/scenario alignment, training-only transforms and fresh CSV examples. All 39 unit tests pass; dashboard startup and detected/missed/false-alarm replays were checked.

Step	Command or artifact
Prepare and freeze	python scripts/prepare_hvac.py
Fit fixed candidates	python scripts/train_hvac.py
Verify validation	python scripts/verify_hvac.py --validation-only
Frozen evaluation	python scripts/evaluate_hvac.py
Verify saved final scores	python scripts/verify_hvac.py
Dashboard	streamlit run dashboard/app.py
Notebook / model card	HVAC_submission.ipynb / results/hvac_v1/MODEL_CARD.md

Project contribution: source/label audit, a controlled comparison, causal preprocessing, independent-season evaluation, seed repetition, explicit false-alarm constraints, retained failure cases and a reproducible demo. The preserved M100 investigation documents why physical telemetry did not solve its tested scheduler-state task; its failed recovery is not advertised as accurate cooling detection.

[1] Lin, G. and Mitchell, R. (2019). Data Sets for Evaluation of Building Fault Detection and Diagnostics Algorithms. Lawrence Berkeley National Laboratory, OEDI. DOI 10.25984/1824861. Official data record.

[2] Granderson, J. and Lin, G. (2019). Inventory of Data Sets for AFDD Evaluation. Source inventory supplied with the collection, pp. 3-7 for MZVAV-1. Local copy: data/hvac_benchmark/source_inventory.pdf.

[3] Dataset authors, M100 ExaData preparation and plugin definitions. Label query; DOI 10.1038/s41597-023-02174-3. Supporting-study details are in results/task5_recovery_v1/LABEL_AND_SENSOR_AUDIT.md.

Exact Python/ML package versions and model/protocol fingerprints are stored with the experiment. Source outcomes and original artifacts remain preserved. No post-test tuning occurred.